

■ Lab 6 — Locals (AWS CloudShell)

■ Objectifs du Lab

- Déclarer des **locals** pour centraliser les expressions
- Référencer avec `local`.
- Utiliser les locals pour les tags communs et les noms de ressources

■ Étape 1 — Créer les fichiers

```
bash $HOME/install-terraform.sh
cd $HOME/terraform-training/lab6-locals
```

```
cat > backend.tf << 'EOF'
terraform {
  backend "s3" {
    region      = "eu-west-1"
    use_lockfile = true
    encrypt     = true
  }
}
EOF

cat > versions.tf << 'EOF'
terraform {
  required_version = ">= 1.15.0"
  required_providers {
    aws = { source = "hashicorp/aws", version = "~> 5.0" }
  }
}
EOF

cat > provider.tf << 'EOF'
provider "aws" { region = var.region }
EOF

cat > variables.tf << 'EOF'
variable "username" { type = string }
variable "region" { type = string; default = "eu-west-1" }
variable "environment" {
  type      = string
  default   = "dev"
  validation {
```

```

        condition      = contains(["dev", "prod"], var.environment)
        error_message = "L'environnement doit être dev ou prod."
    }
}
EOF

cat > locals.tf << 'EOF'
locals {
    prefix          = "lab6-${var.username}-${var.environment}"
    instance_type   = var.environment == "prod" ? "t2.small" : "t2.micro"
    common_tags = {
        Lab          = "lab6"
        Username      = var.username
        Environment   = var.environment
        ManagedBy     = "Terraform"
    }
}
EOF

cat > main.tf << 'EOF'
resource "aws_security_group" "lab6_sg" {
    name          = "${local.prefix}-sg"
    description   = "Security group ${local.prefix}"
    tags          = merge(local.common_tags, { Name = "${local.prefix}-sg" })
}

resource "aws_instance" "lab6_instance" {
    ami           = "ami-0694d931cee176e7d"
    instance_type = local.instance_type
    vpc_security_group_ids = [aws_security_group.lab6_sg.id]

    tags          = merge(local.common_tags, { Name = "${local.prefix}-instance" })
}
EOF

cat > outputs.tf << 'EOF'
output "instance_id" { value = aws_instance.lab6_instance.id }
output "instance_type_used" { value = local.instance_type }
output "prefix_used" { value = local.prefix }
EOF

```

■ Étape 2 — Déployer et observer

```

terraform init
terraform apply -var="username=<votre-prenom>" -var="environment=dev"
terraform output instance_type_used

```

■ Étape 3 — Tester le local conditionnel

```
terraform apply -var="username=<votre-prenom>" -var="environment=prod"
terraform output instance_type_used
```

■ Étape 4 — terraform console

```
terraform console -var="username=<votre-prenom>" -var="environment=dev"
```

```
> local.prefix
> local.instance_type
> local.common_tags
```

Tapez `exit` pour quitter.

■ Étape 5 — Nettoyage

```
terraform destroy -var="username=<votre-prenom>" -var="environment=dev"
```

■ Validation du Lab

#	Critère	Validé
1	`locals.tf` créé avec `prefix`, `common_tags`, `instance_type`	■
2	`instance_type_used` retourne `t2.micro` en dev	■
3	`instance_type_used` retourne `t2.small` en prod	■
4	`terraform console` affiche les valeurs des locals	■
5	`terraform destroy` supprime les ressources	■

■ ****Fin du Lab 6 !****